## 5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```python
'June', 'July', 'August', 'September', 'October', 'November',
'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12,
8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5,
3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}

# Write your code...
df = pd.DataFrame(data)
plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'],df['City_C_Temperature'],
              labels=['City A', 'City B', 'City C'])

plt.xlabel('Month')
plt.ylabel('Temperature')
plt.title('Temperature Variation')
plt.show()
```

Terminal Test cases

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**5.2.1. Titanic Dataset** 00:35

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

### Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

### Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

### Visualization Details:

Write the code to create a series of visualizations as follows:

titanicDat... Submit

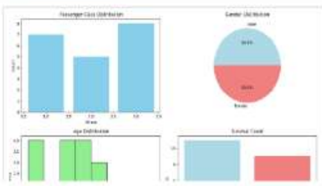
```
# import pandas as pd
# import matplotlib.pyplot as plt

# # Load the Titanic dataset from the CSV file
# df = pd.read_csv('titanic.csv')

```

Average time 1.866 s 1866.00 ms | Maximum time 1.866 s 1866.00 ms

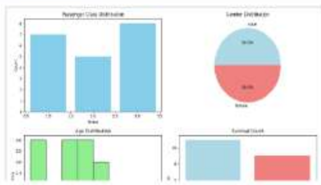
1 out of 1 shown test case(s) passed

Test case 1 1866 ms Debug

Expected output



Actual output



Terminal | Test cases

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**5.2.2. Histogram of passenger information of Titanic** 10:40

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black** (k).
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

```
PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
```

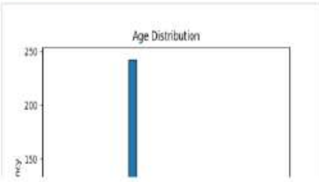
Explorer | Histogra... | Submit | Debugger

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time **0.588 s** 588.00 ms | Maximum time **0.588 s** 588.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 588 ms Debug

Expected output




Actual output


Terminal | Test cases

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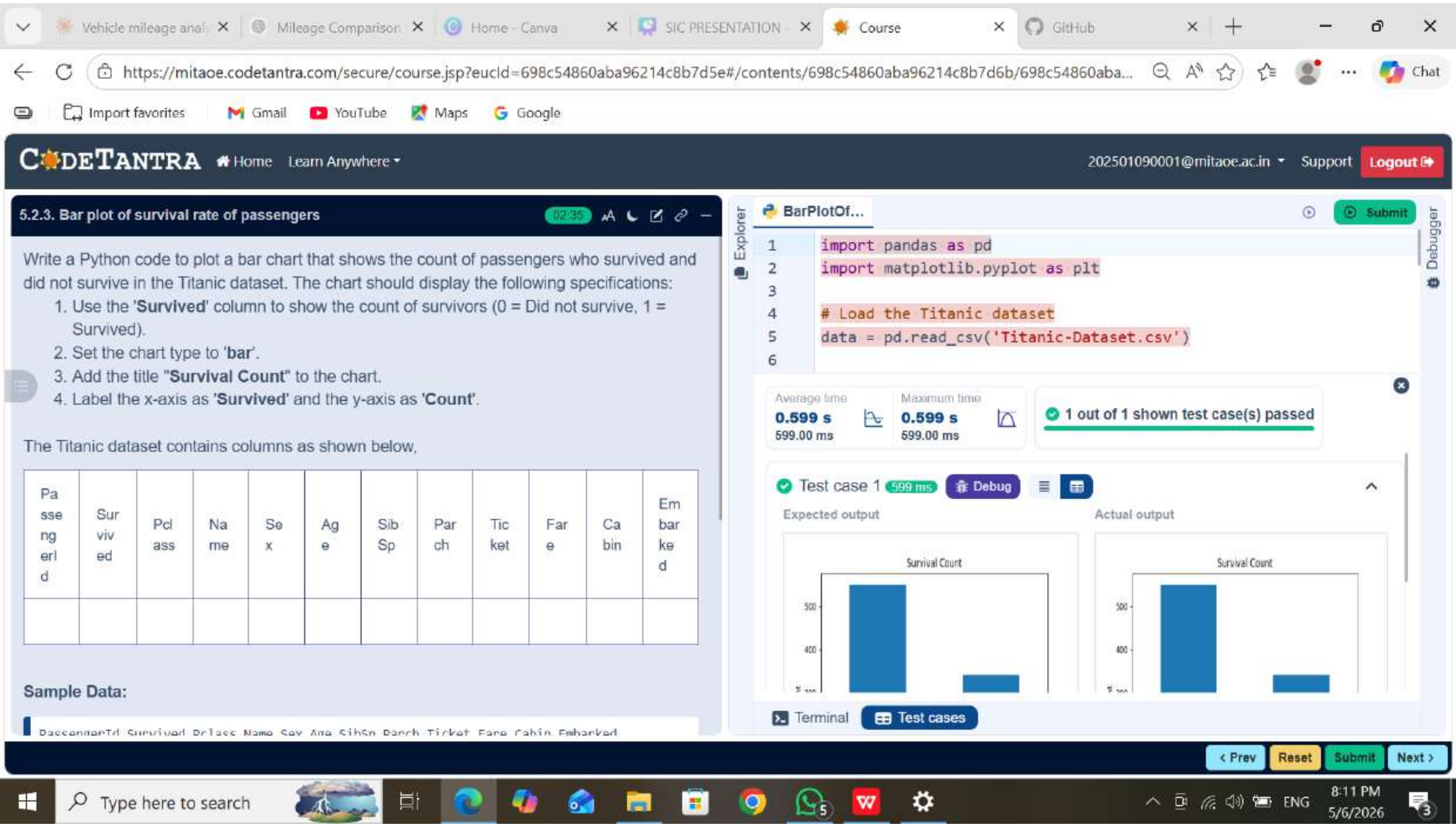
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5.2.3. Bar plot of survival rate of passengers
02:35
Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:
1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title "Survival Count" to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.
The Titanic dataset contains columns as shown below,
Passengerld Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
Sample Data:
PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
Explorer
BarPlotOf...
Submit
Debugger
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
Average time
0.599 s
599.00 ms
Maximum time
0.599 s
599.00 ms
1 out of 1 shown test case(s) passed
Test case 1
599 ms
Debug
Expected output
Actual output
Survival Count
500
400
Survival Count
500
400
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Test cases
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### 5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

BarPlotOf...

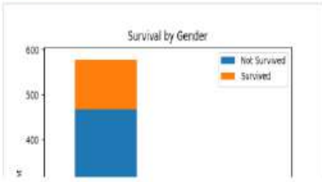
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time 0.681 s 681.00 ms | Maximum time 0.681 s 681.00 ms

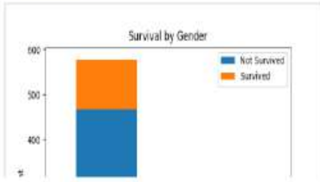
1 out of 1 shown test case(s) passed

Test case 1 681 ms Debug

Expected output




Actual output

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**5.2.5. Bar Plot for Survival by Pclass** 16:34

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

Explorer | BarPlotOf... | Submit | Debugger

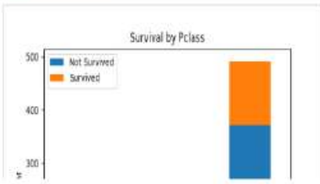
```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time **0.728 s** 728.00 ms | Maximum time **0.728 s** 728.00 ms

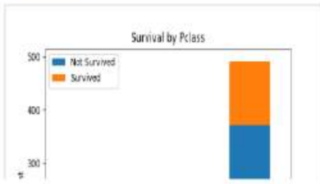
**1 out of 1 shown test case(s) passed**

Test case 1 728 ms | Debug

Expected output




Actual output




Terminal | Test cases

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## 5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked "** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|

BarPlotOf...

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

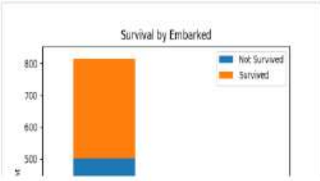
Average time **0.682 s** 682.00 ms

Maximum time **0.682 s** 682.00 ms

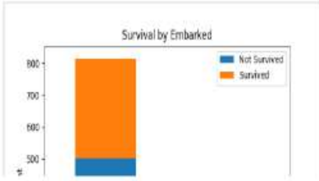
1 out of 1 shown test case(s) passed

Test case 1 682 ms Debug

Expected output




Actual output




Terminal Test cases

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## 5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below.

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

BoxPlotF...

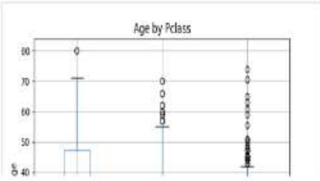
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time 0.678 s 678.00 ms | Maximum time 0.678 s 678.00 ms

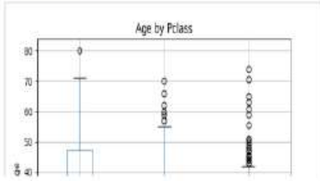
1 out of 1 shown test case(s) passed

Test case 1 678 ms Debug

Expected output




Actual output

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**5.2.8. Box Plot for Age by Survived**  03:17

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

| Passengerld | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

Explorer  BoxPlotF...  Submit  Debugger

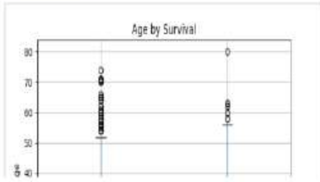
```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time 0.678 s 678.00 ms | Maximum time 0.678 s 678.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 678 ms  Debug

Expected output


Actual output




Terminal  Test cases

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## 5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

```
PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
```

BoxPlotF...

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

```

Average time 0.651 s 651.00 ms

Maximum time 0.651 s 651.00 ms

1 out of 1 shown test case(s) passed

Test case 1 651 ms Debug

Expected output


Actual output


Terminal Test cases

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## 5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
```

AgeFareS... Submit

```
# import pandas as pd
# import matplotlib.pyplot as plt

# # Load the Titanic dataset
# data = pd.read_csv('Titanic-Dataset.csv')

```

Average time **0.538 s** 538.00 ms | Maximum time **0.538 s** 538.00 ms

**1 out of 1 shown test case(s) passed**

Test case 1 538 ms Debug

Expected output


Actual output


Terminal | Test cases

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**5.2.11. Scatter Plot for Age vs. Fare by Survived** 00:28

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**Sample Data:**

Explorer

AgeFareS... Submit

```
# import pandas as pd
# import matplotlib.pyplot as plt

# # Load the Titanic dataset
# data = pd.read_csv('Titanic-Dataset.csv')

```

Debugger

Average time **0.598 s** 598.00 ms

Maximum time **0.598 s** 598.00 ms

1 out of 1 shown test case(s) passed

Test case 1 598 ms Debug

Expected output


Actual output


Terminal | Test cases

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Type here to search

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